

Risks and Responsibilities

Liability

- A society can legally impose liability for actions based on a variety of criteria. The following discussion will address three resulting categories of liability:
 - contract liability
 - criminal liability
 - tort liability

Computer liability

- In a society that depends on computers to do such things as monitor a hospital patient's well-being, assure our national defense and report on the cash flow of businesses, an important issue relates to who should bear the financial loss when computer fails to work properly.
- Computer may malfunction due to :
 - Software malfunction
 - Hardware malfunction
 - Network failure
- Effects of computer errors can cause
 - Inconvenience
 - Bad business decisions
 - Fatalities

Software failure

- ‘Software failure’ has been defined as
 - the occurrence of either deficient functionality, where the program fails to perform a required function, or
 - deficient performance, where the program performs a function too slow or in an insufficient manner.
- Failure may also occur when software is not
 - compatible with other software or
 - it is difficult to maintain or upgrade due to insufficient care by the designer
- Examples:
 - Software billing failure, E-retailer entered wrong price, failure in Aircraft, air traffic control, Nuclear reactor control, Missile systems, Medical treatment systems

Hardware failure

- May be due to
 - incompatible with hardware
 - Insufficient storage
 - Insufficient RAM
 - Improper assembling
 - Very old devices still kept in operation
- examples
 - Rocket launching crash, device not working due to improper assembling, slow processing, frequent restarting or hanged

Engineers' liability

- Engineers must face questions /analyze such as:
 - What can go wrong?
 - How likely is it to happen?
 - What range of consequences might there be?
 - How could they be averted or mitigated?
 - How much RISK should be tolerated or accepted during normal operation?
 - How can risk be measured, reduced and managed?
- Most useful technique
 1. **Identify failure nodes and effects analysis:** identify all the ways a piece of equipment might fail
 2. **Event tree analysis:** identify potential accidents with forward thinking
 3. **Fault tree analysis:** deduction of causes of failure with reverse thinking

Safety in critical system

- Dependability
 - Redundancy
 - Fault tolerance
 - Repairable
 - Maintainable
- Availability
 - Reduce maintenance time, increase dependability
- Safety
 - Primary and secondary; maintain temperature, supply in control,
- Security
 - Access control

Accuracy vs democracy in Internet

Internet – Outreach

The Internet has become the most potent medium of public expression in history



Internet users have the potential to

- share what's on their minds
- express their thoughts
- connect with others or join groups who share their opinions.

Internet – Outreach

The Internet has also become an important medium for the expression of criticism or dissent, especially in socio-political matters.

2012

7 billion worldwide estimated population
2.4 billion active Internet users

Social Media and Free Speech

The rise of Social Media has provided web users with powerful tools of expression.

blogs!

have become the most potent news medium in countries where freedom of the press is curbed.

Social Media and Free Speech

2012

Worldwide



59.4 million Wordpress sites
3.5 billion web pages



87.8 million
Tumblr blogs across the World Wide Web.

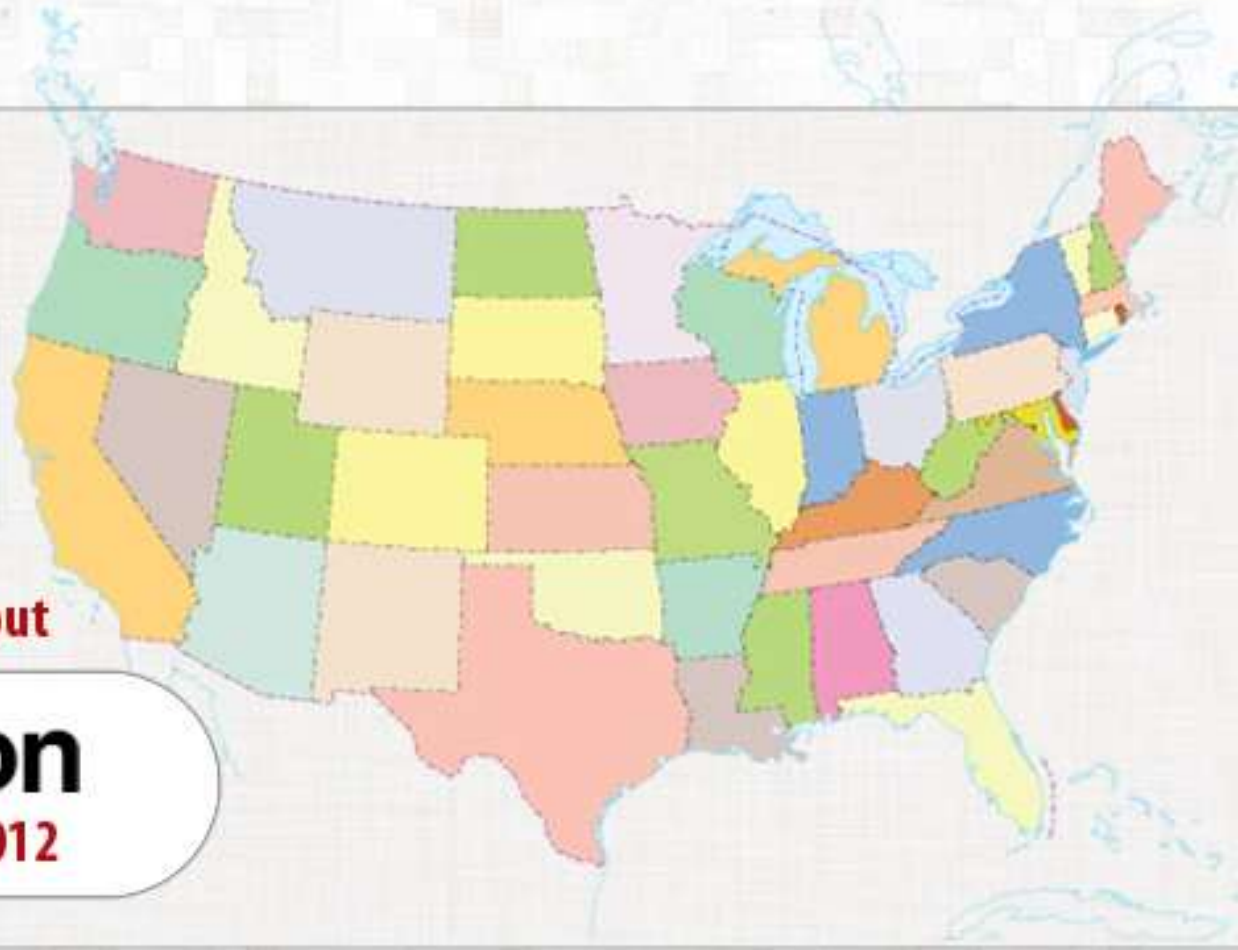
Social Media and Free Speech



In the US there were about

31 million

bloggers as of the end of 2012



Social Media and Free Speech

Apart from these, social networking sites such as



have become important media for voicing one's opinion

OCT
2013



1.5 billion active users
almost a seventh of the world's population

FEB
2013



200 million active users

Social Media and Free Speech

facebook

2012

85,962

Facebook pages posted every month from Brazil alone

MAY

2013

4.75 billion pieces of content were shared daily

Share

every 60 seconds



Status

293,000

statuses are updated



Public

Post

136,000

photos are uploaded

Upload photos

510

comments are posted

Comment

Arab Spring

One of the most significant pro-democratic revolutions of our times was certainly spurred by popular opinions voiced on the Internet.

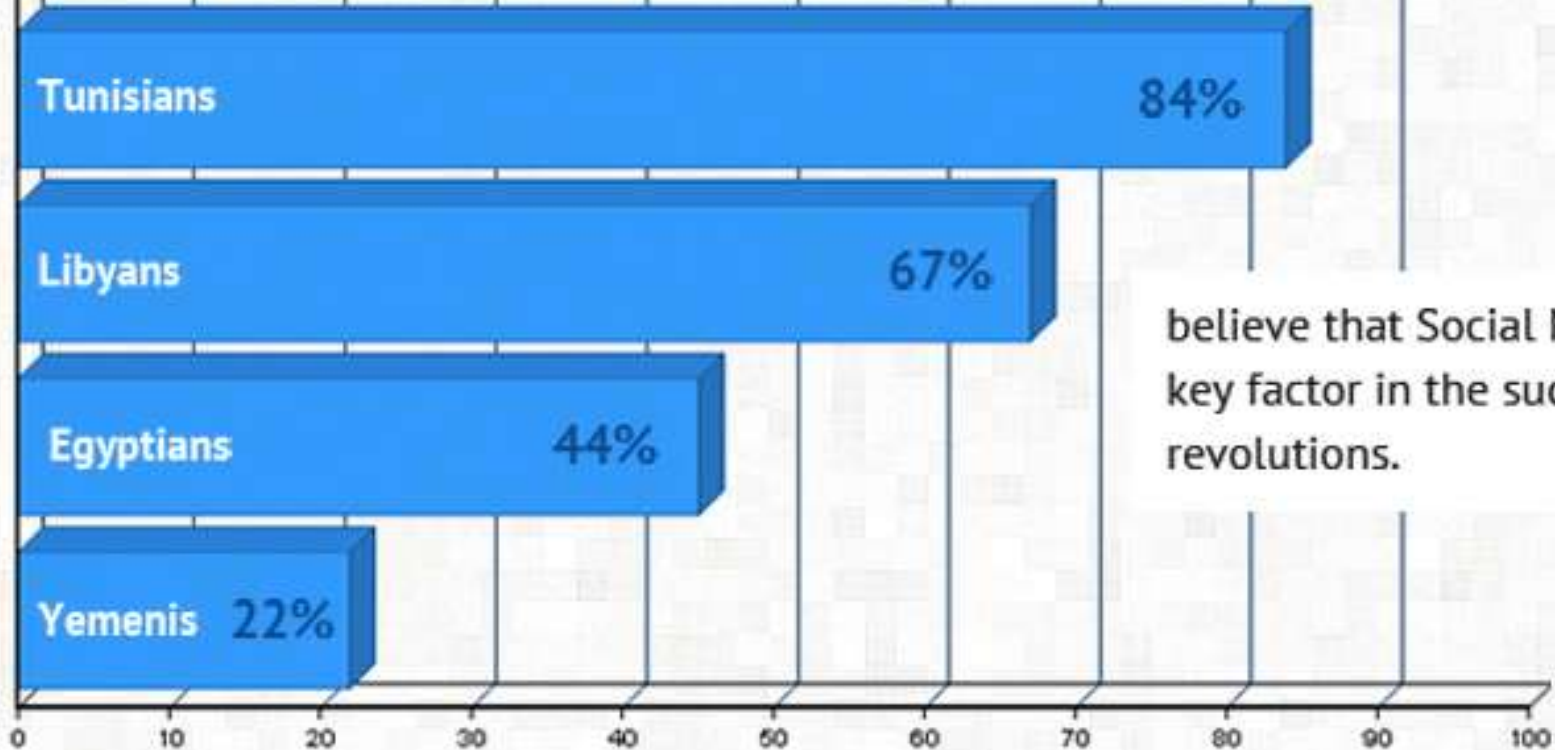


18 DEC
2010

A series of socio-political revolutions broke out in Tunisia and spread like wildfire to many countries of the Arab world.

Arab Spring

According to a survey conducted by Al Jazeera in 2013



believe that Social Media was a key factor in the success of their revolutions.

Accuracy of Information on Websites

No one should assume that information on the Internet is accurate, timely, clear, and important. Many adults have the perception that if something appears in print, then it must be true. Many of our students are even more gullible and are unlikely to know that any person with minimum computer skills can set up a Website. There are no restrictions on what a person can place on a site, and there are no requirements that the material be edited or reviewed. On the one hand, this has significantly broadened the range of information and ideas available to all of us.

The following lists a number of criteria that can be used to evaluate information found on a Website

1. We should consider who recommended the site.
2. the source of the information provided on the site should be evaluated
3. The timeliness of the information should be considered.
4. the mood or “attitude” of the site should be analyzed
5. there should be some attempt to evaluate the accuracy of the information on the site.

Software and design Problem

- **Requirements Volatility**
 - Incomplete or ambiguous requirement collection
- **The Process**
- **Managing Design Influences**
- **Programming**
 - Copy–and–Paste Programming
 - Spaghetti Code
 - Long function
 - Misty variable names
 - Improper documentation

Hardware design issues

- Real time computation
- Size and weight
- Cost effective
- Safe and reliable
- More memory

Elimination of hardware

- For elimination
 - Software centered
 - Online services: cloud computing, e-governance,
 - Distributed systems
 - Wireless technologies
 - Others.....
- Cannot be eliminated completely but reduced down

Professional Responsibilities of Computer Users

- The five areas to be covered under the responsibility of the computer professional are:
 - 1) history of the development and impact of computer technology,
 - 2) why be ethical?
 - 3) major ethical models,
 - 4) definition of computing as a profession, and
 - 5) codes of ethics and professional responsibility for computer professionals.
- Consult Computer Ethics and other code of ethics discussed in Chapter 3 & 4